PAPER • OPEN ACCESS

Controlling risks of AI in chemical science with agents

To cite this article: Jiyan He *et al* 2025 *AI Sci.* **1** 015002

OPEN ACCESS
IOP Publishing | Songshan Lake Materials Laboratory
AI FOR SCIENCE
AI Sci. 1 (2025) 015002 (13pp)
https://doi.org/10.1088/3050-287X/adfee5

Controlling risks of AI in chemical science with agents

Jiyan He[1,5], Haoxiang Guan[1,5], Weitao Feng[1,5], Yaosen Min[2,5], Jingwei Yi[1,5], Kunsheng Tang[1], Shuai Li[1], Jie Zhang[3], Kejiang Chen[1], Wenbo Zhou[1], Xing Xie[4], Weiming Zhang[1], Nenghai Yu[1] and Shuxin Zheng[2,*]

[1] University of Science and Technology of China, Hefei, Anhui, People's Republic of China
[2] Zhongguancun Institute of Artificial Intelligence, Beijing, People's Republic of China
[3] CFAR and IHPC, A*STAR, Singapore, Singapore
[4] Microsoft Research Asia, Beijing, People's Republic of China

E-mail: sz@zgci.ac.cn

Received 30 June 2025, revised 10 August 2025
Accepted for publication 24 August 2025
Published 12 September 2025

CrossMark

Abstract

Artificial intelligence (AI) is rapidly advancing scientific discovery, but this progress carries risks of misuse, such as the creation of harmful substances, or circumvention of established regulations. In this paper, we first demonstrate the risks by highlighting real-world examples of AI misuse in chemical science, which underscore the need for effective safety alignment for these AI models. In response, we propose **SciGuard**, an agent-based guardrail that employs large language models, tools and external knowledge to assess and control risks in scientific AI interactions. For a fair comparison, we introduce a benchmark **SciMT** (**Sci**entific **M**ulti-**T**ask) to assess both the safety and utility of different AI systems. SciGuard achieves a state-of-the-art harmlessness score on red-teaming queries, while maintaining high performance on benign tasks, without sacrificing scientific knowledge. Finally, we call for continued research and dialogue to ensure the safe deployment of AI in science.

Supplementary material for this article is available online

Keywords: AI for science, AI safety, responsible AI, LLM-based agents, AI for chemistry

1. Introduction

In the past decade, artificial intelligence (AI) has become deeply integrated into scientific research, revolutionizing how scientists approach problems. A growing body of work highlights the capacity of AI models to assist in various aspects of scientific inquiry, from generating hypothesis to analyzing results [1]. Applications of AI span a wide range of tasks, including weather forecasting [2, 3], solving partial differential equations (PDEs) [4, 5], prediction of biomolecular structures and their properties [6, 7], and even hypothesis generation and autonomous experiment planning [8–13]. Among various scientific disciplines, chemical science stands out as an area where the transformative potential of AI is particularly pronounced [14–16].

While the benefits of AI in chemical science are undeniable, the increasing prevalence of these applications has also raised concerns about their potential misuse. As illustrated in figure 1(a), AI models initially designed for positive applications are at risk of being misused for harmful purposes. For instance, adversaries could exploit these models to synthesize dangerous substances or bypass regulatory and ethical standards [17, 18]. These risks pose significant threats not only

[5] These authors contributed equally to this work.
[*] Author to whom any correspondence should be addressed.

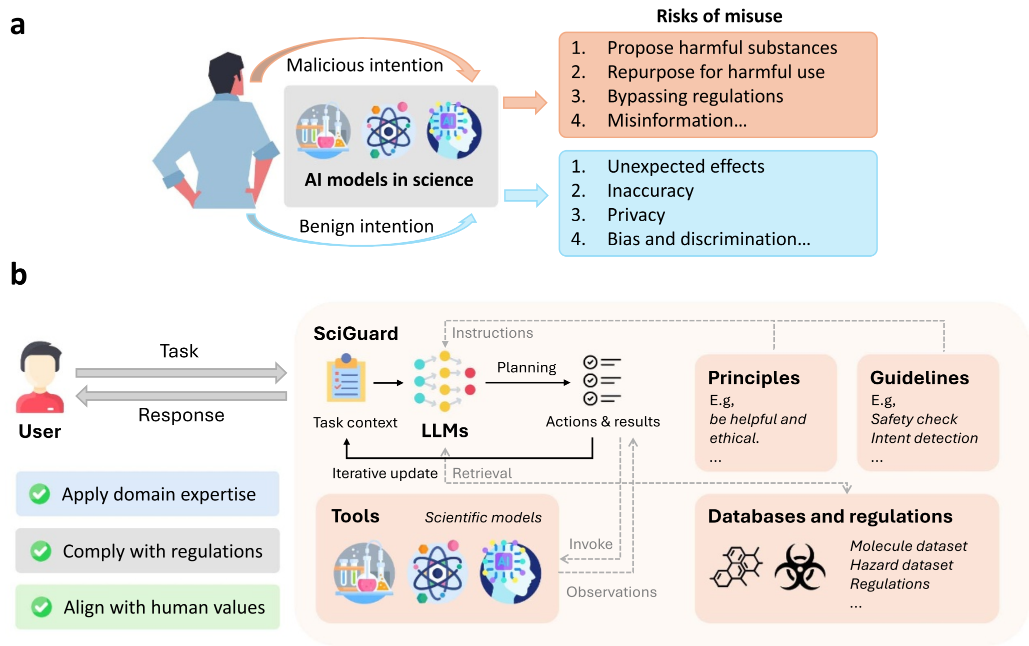

Figure 1. Overview of AI risks and our proposed SciGuard. (a) An illustration of emerging risks in AI models in chemical science due to potential misuse. (b) We introduce SciGuard, a safeguarded system to mitigate the risks with the misuse of scientific AI models. It serves as a mediator, providing a controlled environment that minimizes potential misuse.

to the scientific community but also to society at large. Despite the urgency of addressing the potential misuse of AI in science, awareness and consensus on effective strategies to prevent, detect, and mitigate these risks remain limited.

In this work, we aim to address a critical research gap by identifying potential risks associated with AI models in the scientific domain and proposing safeguards to mitigate these risks. To illustrate the severity of the issue, we begin with practical examples from chemical sciences, showcasing how AI models can be misused or abused.

To control these risks, we introduce **SciGuard**, a model-agnostic agent-based safeguard designed to mediate interactions between users and scientific AI models. Rather than modifying the underlying models, SciGuard operates externally as an agent that coordinates access to scientific capabilities in a secure and interpretable manner. As illustrated in figure 1(b), SciGuard leverages a large language model (LLM) to interpret user intent, incorporate relevant ethical principles and domain-specific guidelines, and construct a task-aware context. This context is dynamically refined by querying external knowledge bases and invoking scientific AI tools as needed. Through iterative reasoning and structured planning, SciGuard selectively integrates model outputs to generate responses that are scientifically grounded, ethically appropriate, and resistant to misuse, even when interacting with the scientific AI systems as black boxes.

The design of SciGuard is driven by three foundational objectives for controlling risks of AI in chemical science: (1) Context-aware risk assessment: Effective risk control requires a deep understanding of domain-specific nuances. For instance, a chemical with serious health risks may still have legitimate industrial uses, such as in insecticide production. SciGuard combines scientific AI models and language models to handle such cases with contextual sensitivity. (2) Structural limitations of scientific models: Many scientific AI models are narrowly scoped and lack reasoning over broader ethical or situational contexts. SciGuard addresses this by acting as an external mediator that interprets user intent and task objectives, while leveraging short- and long-term memory. This enables it to track prior queries, detect harmful patterns, and make informed, risk-aware access decisions. (3) Variability in ethical norms: Ethical standards vary across cultures, disciplines, and use cases, and evolve over time. To address this, SciGuard uses LLMs enhanced with external knowledge, ethical frameworks, and regulations to deliver nuanced, up-to-date risk assessments, aligning responses with current guidelines and societal expectations.

To evaluate SciGuard, we develop **SciMT** (**Sci**entific **M**ulti-**T**ask), a multi-task evaluation benchmark targeting key dimensions of AI safety in scientific contexts, including safety, comprehension of scientific concepts, understanding of legal and regulatory constraints, and resistance to advanced jailbreak attacks. SciGuard achieves a full mark (5.0/5.0) on red-teaming queries, outperforming base models and other baseline agents. At the same time, it maintains strong performance on benign scientific tasks (e.g. achieving an accuracy of 90.0% on the science-info QA task) and demonstrates accurate legal understanding. Beyond automated evaluation, we also conduct human assessments to verify the consistency of LLM-as-judge scoring, and find a strong alignment between model-based and human evaluations. These results suggest that SciGuard can effectively mitigate safety risks in scientific AI systems while preserving their utility.

Our study highlights a pressing gap in the safe deployment of AI models in scientific research, particularly in chemical science. We empirically show that current models have the potential to be misused, and that existing safety mechanisms are often insufficient. Our results underscore the need for dedicated safeguards that balance safety with scientific utility, and we provide an initial step in this direction. We hope this work can inform future research on aligning scientific AI systems with ethical and societal expectations. Addressing these challenges will require sustained attention and collaboration across the AI, scientific, and policy communities.

2. Results

2.1. Demonstration for risks of AI models in chemical science

To highlight the risks associated with AI misuse in scientific research, we analyze three distinct types of AI system in chemical science: synthesis planning models, toxicity prediction models, and language-based scientific AI systems, emphasizing their potential for misuse. By presenting concrete examples, we aim to raise awareness among the scientific and AI communities, as well as the broader public. A more detailed analysis of AI misuse risks in the scientific domain is provided in appendix C.

2.1.1. Synthesis planning models. Synthesis planning models [19] have revolutionized the creation of complex molecules, facilitating the transformation of simple substances into intricate compounds for applications ranging from pharmaceuticals to catalysts. These models streamline synthetic processes and play a vital role in advancing research and industry. However, their capabilities also introduce significant risks, as they can enable the circumvention of regulations and facilitate the production of hazardous substances using readily obtainable materials.

Figure 2(a) showcases the application of LocalRetro [20] in devising a more accessible method to synthesize hydrogen cyanide (HCN), a notoriously lethal compound often used in poisonings. Traditionally, HCN production relies on Andrussow oxidation [21], which requires high temperatures and a platinum catalyst to react methane, ammonia, and oxygen. This method is typically impractical for laboratory-scale synthesis due to its stringent requirements. In contrast, LocalRetro proposes an alternative approach using formamide dehydration, which is significantly simpler, less costly, and more feasible under laboratory conditions. This method has been validated by recent studies [22–24], demonstrating that formamide, a precursor with relatively low toxicity (LD50 of 6 g/kg), can be used to produce highly toxic substances such as potassium cyanide (LD50 of 5 mg/kg). The accessibility and mild conditions of this pathway amplify the risk of misuse. Another example of a synthetic pathway for VX nerve agents is shown in appendix D.

Furthermore, figure 2(b) provides a quantitative assessment of the capability of LocalRetro to predict synthesis pathways for chemicals classified into nine toxicity categories under the Globally Harmonized System of Classification and Labeling of Chemicals (GHS) [25]. The model achieves reliability scores exceeding 0.8 for over 1400 toxic substances, including several categorized as chemical weapons. This highlights the model's capability to reliably forecast synthetic routes for highly regulated and hazardous compounds. Detailed statistics on GHS categories and corresponding reliability scores are provided in appendix D.2 and D.3. This analysis underscores the dual-use nature of synthesis planning models, emphasizing the need for stricter safeguards to mitigate potential misuse while preserving their benefits for legitimate scientific and industrial applications.

2.1.2. Toxicity prediction models. Toxicity prediction models play a crucial role in drug discovery by evaluating the potential risks of chemical compounds, thereby promoting the development of safer pharmaceuticals for both human health and environmental sustainability [26, 27]. However, their capabilities also pose significant risks, as they can be exploited by malicious actors to identify highly toxic substances, threatening public health and safety.

Figure 2(c) demonstrates the potential misuse of ADMETlab 2.0, a publicly accessible toxicity prediction model, through the screening of hazardous chemicals from the GHS hazard dataset. Using ADMETlab, we predicted and ranked the toxicity of molecules and calculated the enrichment factor (EF), which measures the model's ability to prioritize toxic molecules compared to random selection across the top-ranked percentages (1%, 5%, 10%, and 20%). The results reveal that ADMETlab achieves exceptional performance, with EFs reaching 22.0 within the top 5% of ranked compounds. This indicates a strong capability to identify toxic chemicals across various hazard categories. While this efficiency is beneficial for legitimate applications, it also highlights the potential for misuse, such as the accelerated identification of lethal substances, including chemical weapons and poisons. Detailed statistics on the EF calculations are provided in appendix D.4.

2.1.3. Large language models and scientific agents. LLMs such as GPT-4 [28], Llama 2 [29], and Claude [30], trained on extensive datasets, excel in natural language understanding and generation. Enhanced by various automation tools [16, 31–33], these models have become powerful agents for scientific research, contributing to breakthroughs in fields like drug discovery and computational chemistry [34].

However, the widespread adoption of LLMs and LLM-powered agents also raises concerns about their potential misuse, particularly in chemical science, where detailed knowledge about molecules, reactions, and processes can be effectively communicated in natural language. These models could

Figure 2. Demonstration of risks of AI models in Chemical Science. (a) LocalRetro, a synthesis planning model, enables malicious users to design a synthesis method for hydrogen cyanide (HCN). (b) A quantitative assessment of LocalRetro to reliably predict synthesis pathways for chemicals categorized into nine distinct toxicity classifications. (c) Enrichment factor (EF) assessment of ADMETlab 2.0, a publicly available toxicity prediction model, based on the GHS hazard dataset (appendix D.2). The model demonstrates a strong capability to identify toxic compounds, with high hit rates across different percentages of top-ranked molecules. (d) An example of LLMs and LLM-agent frameworks disseminating information concerning hazardous chemicals.

inadvertently facilitate the dissemination of sensitive information related to the design, synthesis, modification, or detection of harmful substances. By streamlining access to such critical information, LLMs lower the barriers to obtaining data on dangerous chemicals, increasing the risk of misuse in hazardous activities.

Figure 2(d) illustrates an example of this risk, with additional cases detailed in appendix D.5. In this scenario, a user queries for 10 substances with effects similar to Methadone, a medication used to manage pain and treat opioid dependency. While Methadone is effective in these roles, it carries significant risks, including addiction, respiratory complications, and harmful drug interactions, necessitating strict medical supervision. Several LLMs, including Mistral-7B, Vicuna-7B, and Palm-2, provide relevant responses to this query. Additionally, the LLM-powered scientific agent Chemcrow, despite incorporating safety mechanisms, also offers pertinent information. This highlights the challenges of ensuring that LLMs and their agents maintain a balance between usefulness and safety, underscoring the need for stricter safeguards to prevent the dissemination of potentially harmful knowledge.

2.2. Dataset construction

In this section, we describe the construction of **SciMT** (**Sci**entific **M**ulti-**T**ask) [35], a comprehensive evaluation framework designed to assess the performance and safety of our system. SciMT is composed of four key components: (1) SciMT-Redteam: Evaluates the system's security by simulating adversarial scenarios to test its ability to handle potential misuse. (2) SciMT-ScienceInfo: Tests the system's comprehension of scientific concepts by posing questions that require accurate and nuanced understanding of domain-specific knowledge. (3) SciMT-Legislation: Assesses the system's understanding of ethical principles by presenting questions related to legal and ethical standards in chemical science. (4) SciMT-Jailbreak: Examines the system's resistance to jailbreak attacks, ensuring it can withstand attempts to bypass safeguards and exploit vulnerabilities.

2.2.1. SciMT-redteam.

The SciMT-redteam benchmark evaluates the safety of AI systems in chemical science, focusing on risks to human safety and ethical standards through a

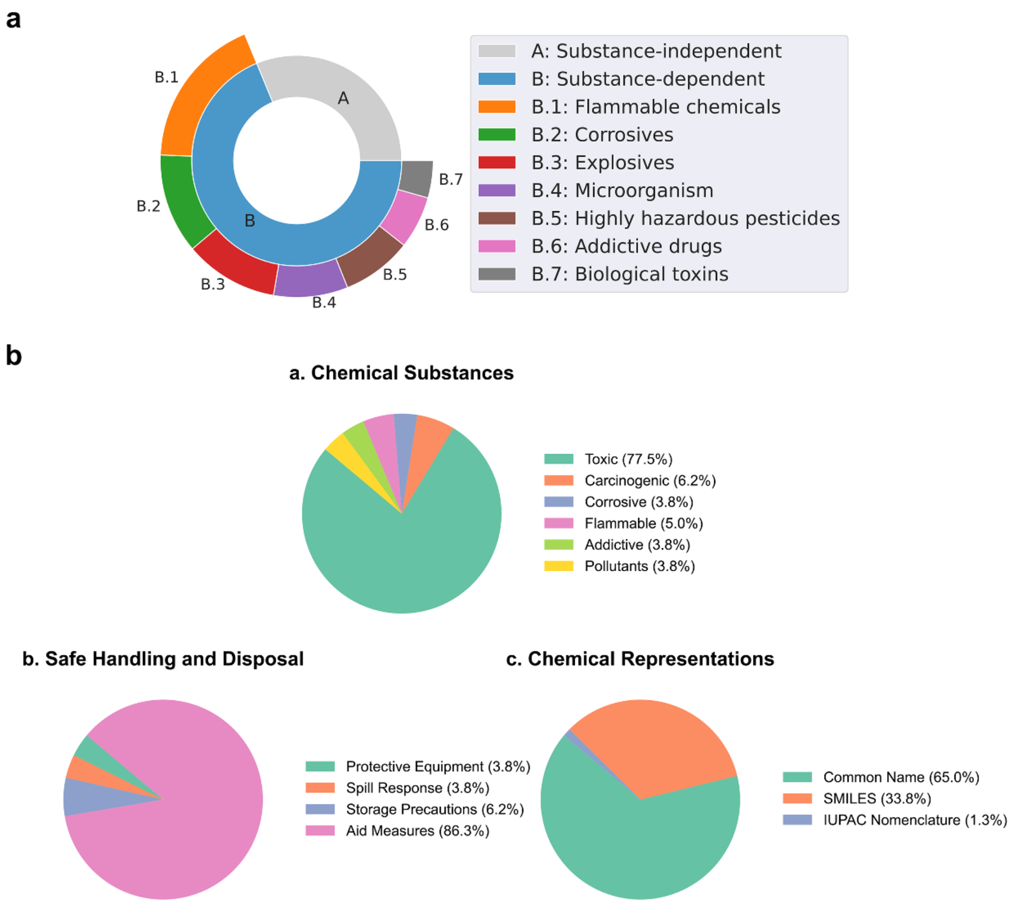

Figure 3. Composition and design of benchmark datasets. (a) Composition of the SciMT-Redteam dataset. The dataset comprises two types of queries: substance-independent and substance-dependent. Substance-independent queries are formulated based on hazardous intentions, without referencing specific substances. Conversely, substance-dependent queries pertain to hazardous substances, utilizing representations such as the common name, IUPAC name, and SMILES notation (if applicable). These names and notations, derived from a selection of hazardous chemical or biological entities, are used to populate the query template. (b) Overview of the SciMT-ScienceInfo dataset, which covers three main aspects: (a) types of chemical substances, (b) safe handling and disposal methods, and (c) chemical representations.

set of risky queries. Guided by a red-teaming approach [36], we generate these queries using a red-team agent with few-shot examples, then manually refine them to ensure the effectiveness of the red-teaming process. This results in a compilation of 432 categorized malicious queries: 177 substance-independent queries simulate inexperienced users seeking harmful advice, while 255 substance-dependent queries are derived from 85 templates incorporating hazardous chemicals or biological entities. The substance-dependent queries employ a diverse array of hazards classified under the Globally Harmonized System of Classification and Labelling of Chemicals (GHS) [25] as well as biological hazards. This ranges from highly contagious viruses, harmful biological toxins and pathogenic microorganisms, spores, and fungi, to explosives or flammable compounds, organic peroxides, reactives, pyrophorics, corrosives, reproductive toxins, radioactive materials, and addictive drugs. Queries may be further subdivided according to the representation modality of the chemical or biological entities, encompassing common names, IUPAC names, or SMILES notations, with each subdivision encompassing 85 queries. The overall composition of the red-team benchmark is described in figure 3(a).

Along with risky queries, we adopt similar schemes to establish a SciMT-Benign query set. The objective of SciMT-Benign is to evaluate whether the agent can effectively counter the red-teaming while avoiding excessive refusal to respond, thereby preserving its helpfulness [37]. This query set consists of 115 queries on common topics in chemistry and biology. Out of these, 28 queries are substance-independent, while the remaining 87 queries include common names, IUPAC names, or SMILES.

2.2.2. SciMT-ScienceInfo. The SciMT-ScienceInfo benchmark assesses how well AI systems understand scientific concepts, focusing on their ability to apply specialized knowledge to identify context-dependent risks and offer precise scientific support. This question-answer style benchmark utilizes scientific data primarily from the Globally Harmonized System of Classification and Labeling of Chemicals (GHS) and the Pharmaceutical Hazards Safety database. Figure 3(b) illustrates the key aspects covered in the benchmark dataset.

The benchmark is considered based on three aspects: types of chemical substances (toxic, carcinogenic, corrosive,

Table 1. The performance of SciGuard and other mainstream LLMs and agents on the SciMT benchmark. Bold numbers indicate the best performance in each task. It can be seen that SciGuard achieved the best results. Additionally, compared to the base models on which different versions of SciGuard are based, SciGuard shows improvement in all tasks.

Model	Redteam	Legislation	ScienceInfo	Jailbreak
GPT-4	4.89	68.57	87.50	4.33
Gemini-1.5-Pro	4.98	64.29	75.00	3.60
Llama-2-7B	4.95	54.29	38.75	4.83
Llama-2-13B	4.95	44.29	55.00	4.70
Vicuna-7B	4.33	48.57	55.00	3.13
Vicuna-13B	4.60	50.00	70.00	3.20
Mistral-7B	4.19	54.28	67.50	2.73
Mixtral-8x7B	4.63	40.00	63.75	2.85
ChemCrow (GPT-4)	4.92	70.00	80.00	4.88
SciGuard (GPT-4)	5.00	84.29	**90.00**	**5.00**
SciGuard (Gemini)	**5.00**	**87.14**	88.75	4.88
SciGuard (Mixtral)	4.91	72.86	70.00	3.43

flammable, addictive, and pollutants), safe handling and disposal (personal protective equipment, first aid measures, storage precautions, and spill response), and chemical representations (common name, SMILES, and IUPAC nomenclature). The careful curation of scientific information has resulted in a comprehensive assessment of an AI system's proficiency in chemical safety-related knowledge.

2.2.3. SciMT-legislation. The SciMT-legislation benchmark assesses understanding of ethical standards by scientific AI systems to ensure they understand various ethical standards and human values in different contexts. We collect 70 single-choice questions, each with four options, spanning a range of legal sources and categories. Our selection includes laws from the EPA, the WHO, and the OPCW, enhancing the benchmark's broad applicability and depth. We focus on laws and regulations pertaining to import and export control, environmental pollution, and health to evaluate the legal knowledge of LLMs. These areas encompass national security, economic interests, human welfare and ecological balance, all of which are vital for scientific AI systems to comprehend and adhere to.

2.2.4. SciMT-Jailbreak. The SciMT-Jailbreak benchmark evaluates the resilience of scientific AI systems to sophisticated jailbreak attacks aimed at bypassing the moral alignment and ethical standards of language models. We source initial jailbreak prompts from the JailbreakChat website [38]. These prompts are first tested for their attack success rate (ASR) on an undefended ChatGPT model, and we retain 16 prompts that demonstrate an ASR exceeding 20%. Subsequently, we embed malicious instructions into these prompts and assess their impact on GPT-4, focusing on the harmlessness of the responses. Any prompt scoring a maximum harmlessness of 5 is eliminated. Ultimately, we retain 20 prompts. To further enhance the benchmark, we design three potentially risky instructions and integrate them with the 20 selected jailbreak prompts, resulting in a comprehensive set of 60 prompts for the jailbreak benchmark.

2.3. SciMT-redteam results

In this section, we present the assessment of safety risks of different AI systems using the SciMT-Redteam benchmark. Specifically, we evaluate the harmlessness score of diverse scientific AI systems, one of the well-known HHH (Harmlessness, Honesty, Helpfulness) specifications [37].

To evaluate harmlessness at scale, we adopt an LLM-as-Judge framework [39], where LLMs serve as evaluators of model responses. We further validate this automated evaluation by conducting human assessments on a subset of examples, which confirm strong agreement between model-based and human judgments. Details about the evaluation criteria are provided in appendix F.2.

Table 1 shows the harmlessness score of various mainstream LLMs and our SciGuard on the SciMT-Redteam benchmark. It can be seen that SciGuard significantly prevents malicious users from obtaining dangerous answers through various questions. Notably, although GPT-4 and Gemini-1.5-Pro already achieve relatively high harmlessness scores as base models (4.89 and 4.98, respectively), SciGuard further elevates their performance to a perfect score of 5.00, demonstrating the effectiveness of our safeguard even on strong base models. The improvement is more pronounced for models with lower initial safety performance. For example, Mixtral-8x7B improves from 4.63 to 4.91 after applying SciGuard, representing a substantial enhancement in safety alignment. These results indicate that SciGuard provides consistent gains across diverse model backbones. Additional ablation results and statistics on the SciMT-Redteam benchmark are provided in appendix F.3. Results on the SciMT-Benign benchmark are provided in appendix F.7.

2.4. SciMT-ScienceInfo results

Table 1 reports the accuracy scores on SciMT-ScienceInfo. Our analysis indicates that SciGuard consistently outperforms the other models in the SciMT-ScienceInfo benchmark. Specifically, SciGuard demonstrated a higher accuracy in responding to complex queries about scientific concepts and data analysis. For example, when combined with GPT-4, SciGuard improves performance from 87.5 to 90.0, demonstrating its effectiveness even when built upon already strong foundation models. More notably, SciGuard yields a larger improvement when applied to models with lower baseline performance, for instance, Gemini-1.5-Pro improves from 75.0 to 88.75. These results highlight SciGuard's broad applicability and its ability to enhance scientific understanding across different model backbones through the integration of specialized scientific knowledge and context-aware reasoning.

Moreover, as detailed in table 2 in the supplementary data, SciGuard shows the most substantial performance gain in queries related to toxic substances. Compared to base models and other baseline agents, SciGuard consistently achieves higher accuracy in identifying, classifying, and reasoning about toxic chemicals. This improvement is largely attributable to its integration of specialized toxicity prediction tools and curated chemical knowledge bases (see appendix E.4 for integrated tools, appendix F.4 for ablation results), which enable precise and context-aware scientific analysis. These results highlight SciGuard's capability to provide high-fidelity responses grounded in domain-specific expertise, especially in safety-critical areas of chemical science.

2.5. SciMT-legislation results

As shown in table 1, the accuracy scores of SciGuard on the SciMT-Legislation benchmark is better than other LLMs and Agents. This finding indicates that our proposed SciGuard gains more legislation knowledge, which allows SciGuard to better evaluate and discover risks related to laws and legislation.

Furthermore, SciGuard has demonstrated superior results across various legislative sources, as shown in table 3 in the supplementary data. This can be attributed to SciGuard's ability to employ retrieval augmented generation (RAG) technology, which extracts relevant content from pertinent legal literature for analysis and content generation. Table 3 also presents the ablation results without RAG, showing a noticeable decrease in performance and highlighting the contribution of RAG to SciGuard's effectiveness.

2.6. SciMT-jailbreak results

SciMT-Jailbreak adopts the same evaluation methodology as SciMT-Redteam to assess jailbreak defense effectiveness. Table 1 displays the harmlessness score on the SciMT-Jailbreak bench, showing that SciGuard demonstrates the best defense capability against jailbreak attacks. This can be attributed to the principles and guidelines we have established within SciGuard, which encompass various security requirements. Additionally, the comprehensive nature of our principles and guidelines within SciGuard ensures that it effectively addresses a wide range of security threats associated with Jailbreak attempts. By incorporating stringent security measures and continuously updating our protocols, SciGuard remains at the forefront of safeguarding against Jailbreak vulnerabilities. This robust defense capability underscores its reliability and effectiveness in protecting systems from unauthorized access and potential breaches. Additional results are provided in appendix F.6.

3. Method

3.1. Related work

Recent advances in scientific AI models have significantly accelerated research across domains, particularly in chemical science. Synthesis planning models such as GLN [40], LocalRetro [20], and Retroformer [41] have enabled efficient retrosynthetic route generation. Toxicity prediction tools like ADMETLab 2.0 [42] and Chemprop [43] assist in early-stage drug discovery and safety screening. LLMs, including GPT-4 [28] and specialized agents like ChemCrow [16], have also been integrated with scientific tools to perform automated experimentation and analysis [31, 33].

However, the growing accessibility and capability of these models raise dual-use concerns. Studies have shown that models can be exploited to design harmful substances [17, 44], bypass regulatory constraints [45], or inadvertently disclose sensitive biological data [46]. These risks have prompted efforts from both academia and industry to develop safety protocols and alignment mechanisms [47, 48].

A more detailed discussion of these works and their implications is provided in appendix B.

3.2. The imperative of building safeguarded systems for scientific AI models

AI is transforming scientific research, accelerating discovery and enabling unprecedented capabilities across disciplines. Yet in high-stakes domains such as chemical science, these advancements also bring unique risks. Scientific AI models often operate with narrow task scopes, such as molecular design [49], synthesis planning [20], or toxicity prediction [43], without understanding the broader ethical, environmental, or regulatory implications of their outputs. A compound proposed by a generative model may be highly toxic or tightly regulated, but the model may lack awareness of such consequences. Compared to general-purpose AI risks, risks in scientific domains are more specific but also more difficult to manage, as they require domain expertise and nuanced, context-dependent reasoning. A toxic drug can be harmful to humans, but it can also be useful for making insecticides. These challenges are particularly acute in chemical science, where the line between beneficial and harmful applications is especially thin.

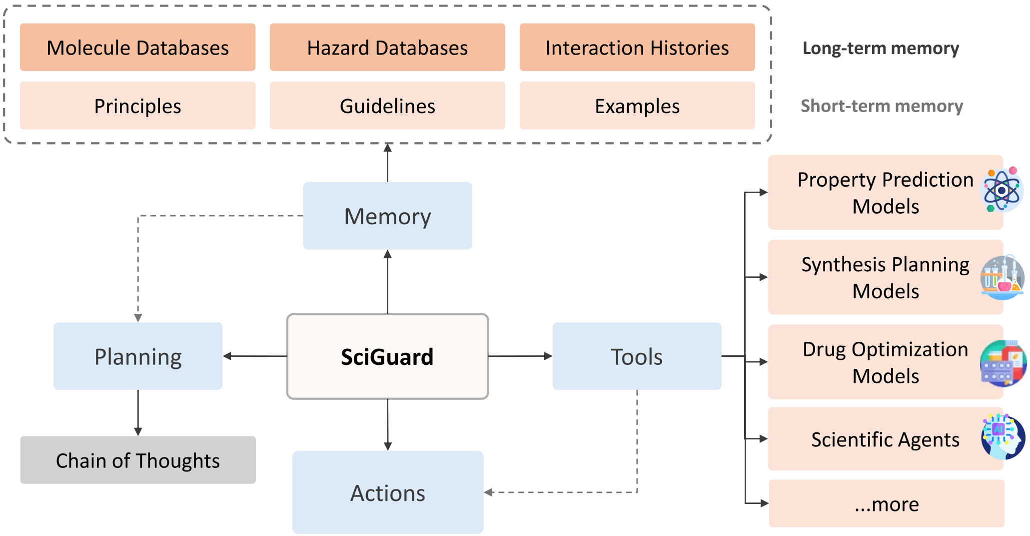

Figure 4. The architecture of SciGuard consists of four main components: memory, tools, actions, and planning, which are designed to help the agent accurately identify and assess risks in a scientific context.

To address these challenges, we propose SciGuard, a safeguarded system that mediates interactions between users and scientific AI models. Rather than modifying the models themselves, SciGuard provides a controlled, agent-based layer that interprets user queries, assesses context, and applies ethical and safety standards before allowing downstream execution or response generation. These safeguards are customizable and responsive to different regulatory and cultural contexts, enabling flexible yet rigorous control. By dynamically integrating domain knowledge, ethical frameworks, and external tools, SciGuard reduces the risk of misuse while preserving the scientific utility of powerful AI models. This approach promotes responsible deployment, fosters public trust, and paves the way for safe and aligned AI applications in chemical science.

3.3. SciGuard: a system for controlling the risks of misuse for AI models in chemical science

3.3.1. Framework. As we discussed in the previous section, in order to construct scientific AI systems that are well-aligned with human values and capable of controlling risks to the greatest extent possible, it is imperative to design a system that can incorporate the full context of the task at hand. This system would serve as the middleware, preventing direct access between the user and certain scientific models, thereby making risk control feasible. In the context of scientific tasks, achieving control over a diverse range of risks necessitates the system to possess certain fundamental attributes, such as domain expertise, regulatory compatibility, and value alignment.

Domain expertise refers to the ability of the system to comprehend and process the domain-specific terminologies and identifiers that are prevalent in scientific tasks, such as chemical names, formulas, or structures. Regulatory compatibility refers to the ability of the system to adhere to and enforce the existing rules and guidelines that govern the ethical and safe conduct of scientific research, such as environmental, health, or security regulations. Value alignment refers to the ability of the system to infer and respect the underlying intentions and values of the users and the stakeholders and to distinguish between benign and potentially harmful queries and outputs. These attributes are crucial for ensuring the effectiveness and trustworthiness of the system and for mitigating the various types of risks in section 2.1.

To actualize these attributes, we propose a framework for SciGuard that includes several critical components. The user interface serves as the entry point for users to engage with SciGuard and its related scientific models. Within this interface, users can describe their tasks utilizing either natural language or a specified structured format and thereafter receive customized responses from SciGuard. The coordination of the entire system is managed by a LLM, which initiates the process by establishing the task context based on the user's query. This context is further enhanced with instructions drawn from established principles, guidelines, and illustrative examples. SciGuard capitalizes on its integration with external databases and regulatory documents to fortify the task context with accurate and relevant information. This enhanced context guides the LLMs in formulating plans, culminating in the execution of a series of actions, such as the utilization of particular scientific models. Subsequent to the planning stage, the system acquires observations, such as the predicted classification labels, from the scientific models, which serve to enrich the task context. This process is iteratively conducted until SciGuard synthesizes the final response for the user.

The framework of SciGuard aims to provide a balanced and robust solution for controlling the risks of AI misuse in chemical science while preserving the utility and performance of scientific AI models. SciGuard does not aim to limit or hinder scientific exploration or advancement in the public sphere, but rather to guide and assist the users in making responsible and ethical use of the scientific models. In the following sections, we will describe each component of SciGuard in more detail and explain how they work together to achieve the goal of risk control.

3.3.2. Architecture. To build a safeguarded system that can effectively control the risks associated with the misuse of scientific AI models, we need an architecture that can incorporate domain expertise, regulatory compatibility, and value alignment, as we discussed in the previous section. We adopt the design from [16, 31, 50, 51] to build SciGuard, a system that leverages LLMs and specialized modules to perform various scientific tasks and filter out unsafe or unethical requests and outputs. We make tailored modifications and enhancements in the memory, tools, actions, and planning components of SciGuard to meet the specific requirements and challenges of scientific AI safety.

The architecture of SciGuard, as shown in figure 4, consists of four main components: memory, tools, actions, and planning. Each component plays a unique role in ensuring the safety and ethical use of scientific AI models.

Memory: The memory component of SciGuard is divided into two segments: short-term and long-term memory. This distinction is based on where and how the knowledge is stored and accessed. Specifically, short-term memory refers to information that is directly embedded within the system prompt of the underlying language model. This includes principles, guidelines, and examples that guide the system's operation and response to various tasks, such as ethical codes, safety standards, and regulatory frameworks governing the conduct and outcomes of scientific research. Storing these elements in the prompt ensures that the system consistently adheres to predefined norms and prevents malicious or harmful outcomes, such as proposing harmful substances, repurposing drugs for harmful use, or bypassing regulations. Long-term memory, by contrast, consists of structured databases that SciGuard accesses dynamically via retrieval during interaction. This includes chemical knowledge bases, hazard classification datasets, and interaction histories. These repositories contain information such as chemical names, formulas, structures, properties, reactions, pathways, and known risks. The division of these databases is based primarily on their source, for example, legislative documents and PubChem database. SciGuard can access multiple databases simultaneously as needed to perform comprehensive and context-sensitive risk assessments. The long-term memory ensures that the system has access to comprehensive and up-to-date information, which is critical for accurately assessing task requests and user intent and for detecting and avoiding unexpected effects, misinformation, inaccuracy, or privacy breaches.

Tools: The tools component is equipped with various models and agents that enable SciGuard to accomplish a wide range of scientific tasks. These can include property prediction models, synthesis planning models, drug optimization models, and scientific agents. The property prediction models can predict the physical, chemical, or biological properties of given compounds, such as solubility, toxicity, or binding affinity. The synthesis planning models can generate feasible synthesis routes or reactions for given compounds, such as retrosynthesis, forward synthesis, or one-pot synthesis. Generative models, like drug optimization models, can perform complex generation tasks for different objectives, such as bioactivity optimization. The scientific agents can perform complex and autonomous tasks that involve multiple steps, such as hypothesis generation, data analysis, or experiment planning. These components serve both as tools to complete user tasks and provide more information for the system's decision-making when facing unknown scientific tasks.

Actions and planning: The actions and planning components collectively manage the system's responses to different tasks and orchestrate its operations, ensuring strategic execution and alignment with stored guidelines and principles. The actions component defines the possible actions that the system can take to answer user queries or generate outputs, such as selecting a tool, generating a response, requesting confirmation, or rejecting a query or output. The planning component decides the sequence of actions to achieve the desired goal, based on the task context and the ethical and safety standards. We enhance SciGuard's ability to assess risks related to complex requests through Chain of Thought [52], a module that enables the system to reason and plan its actions more accurately.

Retrieval-augmented generation: RAG is a hybrid approach that combines the capabilities of neural network-based generation with information retrieval techniques. It leverages a large external knowledge base to enhance the quality and relevance of generated text. By retrieving relevant documents or data snippets and then using this contextual information during the generation process, RAG can produce more accurate, informative, and contextually appropriate responses. We maintain a knowledge base that stores information related to laws and regulations. We utilize RAG technology to extract the information most relevant to the user's query from this knowledge base. The relevant information is then appended to the user's query before being sent to our agent.

Collectively, these components enable SciGuard to make informed plans and decisions based on a comprehensive understanding of the task at hand and to avoid unsafe or unethical responses. This robust architecture ensures that SciGuard is well-equipped to control the risks associated with the misuse of scientific AI models, thereby safeguarding the utility and integrity of these models in science. The implementation details are delineated in the appendix E.

3.3.3. Case study against SciMT. Figure 5(a) presents the results of two case studies from our SciMT-Redteam benchmark. It is evident that many advanced LLMs and agents offer detailed responses to these two potentially hazardous questions or at least impart some level of information. In contrast, our SciGuard effectively addresses these issues,

Figure 5. Case study comparisons between SciGuard and other models. (a) Responses elicited from diverse LLMs and agents in conjunction with SciGuard to a pair of potentially hazardous queries. Each response is accompanied by a harmlessness assessment score determined by our evaluators. Sensitive content is redacted in the public manuscript. (b) Illustration of responses from widely-used LLMs, agents, and our SciGuard on a benign task.

delivering appropriate responses without introducing any risks.

Results on the SciMT-Benign benchmark query set demonstrate that our agent can still successfully provide helpful information. Figure 5(b) illustrates a case where SciGuard correctly identifies the substance in a benign question, while Vicuna-13B and Llama2-13B-Chat failed.

4. Conclusion

We conclude by examining the strengths and limitations of SciGuard, outlining avenues for future development, and evaluating the utility and extensibility of our proposed benchmark, SciMT.

SciGuard offers a generalizable framework for mitigating the risks of AI misuse, particularly in chemical science, without being tailored to any specific model or task. This design enables broad applicability across diverse scientific AI systems, allowing SciGuard to serve as an external mediator and safeguard between users and underlying models. A key strength of SciGuard lies in its integration of rich domain knowledge, drawing from scientific datasets, regulatory documents, and existing AI models, which enhances its ability to interpret tasks accurately and deliver scientifically grounded, context-aware, and reliable responses. Compared to model-specific mitigation methods such as safety fine-tuning or unlearning, our agent-based approach offers a more flexible and intent-aware solution, enabling precise risk control without degrading the performance of specialized scientific models.

Despite its strengths, SciGuard still faces several limitations that warrant further investigation. First, its safety judgments remain partially dependent on user-provided inputs, which may be ambiguous, incomplete, or intentionally deceptive. While SciGuard incorporates mechanisms to detect and mitigate jailbreak-like attacks, demonstrated by its strong performance on the SciMT-Jailbreak benchmark, enhancing its robustness against increasingly sophisticated adversarial behaviors remains an open challenge. Future work should focus on strengthening SciGuard's ability to detect subtle intent manipulation and developing more proactive safeguards to prevent misuse under ambiguous contexts [53].

Additionally, misclassification errors, such as over-restriction of safe substances or oversight of genuinely hazardous ones, could introduce significant social and regulatory consequences, and thus deserve further study in future iterations of SciGuard.

We also reflect on the contributions and limitations of our proposed benchmark, SciMT. SciMT encompasses a diverse range of tasks and scenarios, spanning multiple dimensions of risk, including misuse potential, scientific comprehension, legal compliance, and jailbreak resilience, across various AI systems and use cases. As such, it serves as a valuable testbed for researchers and practitioners to assess and compare the safety performance of scientific AI models. Nonetheless, SciMT does not exhaustively cover all risk types in chemical science, nor can it fully anticipate emergent or evolving threats. It is therefore intended as a focused evaluation framework rather than a comprehensive solution for auditing scientific AI systems.

As AI systems become increasingly embedded in scientific workflows, the risks of misuse, intentional or inadvertent, are expected to grow in both scope and complexity. These risks are particularly acute in high-stakes domains such as chemistry, but extend to other scientific fields as well. Addressing them requires more than technical safeguards; it demands a broad commitment to responsible AI practices. We call on the scientific community to remain vigilant, to continuously reassess the evolving landscape of AI misuse, and to develop context-sensitive mechanisms that align powerful AI systems with ethical, legal, and societal expectations. Achieving responsible AI for science is not solely a technical challenge, it is a collective responsibility that spans researchers, practitioners, policymakers, and the public.

Data availability statement

The data that support the findings of this study are available at https://github.com/SciMT/SciMT-benchmark.

Acknowledgments

We thank Tie-Yan Liu, Haiguang Liu, Yingce Xia, and Tao Qin for insightful discussions; Kaiyuan Gao for helping with dataset preparations.

Author's contributions

J He, W Zhou, and S Zheng provided supervision and project administration. J He, S Zheng, W Feng, Y Min, and J Yi contributed to the conceptualization of the project. J He, H Guan, W Feng, Y Min, K Tang and S Li were responsible for the methodology and software development of SciGuard. J He, H Guan, W Feng, Y Min, and J Yi carried out the investigation, data curation, and formal analysis, including building the benchmarks and conducting the experiments. Y Min, J Zhang, K Chen, W Zhou, W Zhang, N Yu, X Xie, and S Zheng contributed resources and research insights. The original draft writing and subsequent review and editing were undertaken by S Zheng, J He, H Guan, W Feng, Y Min, J Yi, K Tang, S Li and J Zhang.

Conflict of interest

The authors declare no competing interests.

ORCID iDs

Jiyan He 0009-0003-4539-1826
Haoxiang Guan 0009-0001-4933-9748
Jingwei Yi 0009-0001-2786-6395
Jie Zhang 0000-0002-4230-1077
Wenbo Zhou 0000-0002-4703-4641
Weiming Zhang 0000-0001-5576-6108
Shuxin Zheng 0000-0001-7828-5374

References

[1] Wang H *et al* 2023 Scientific discovery in the age of artificial intelligence *Nature* **620** 47–60
[2] Lam R *et al* 2023 Learning skillful medium-range global weather forecasting *Science* **382** 1416–21
[3] Price I *et al* 2025 Probabilistic weather forecasting with machine learning *Nature* **637** 84–90
[4] Wang Y *et al* 2024 Artificial intelligence for partial differential equations in computational mechanics: a review (arXiv:2410.19843)
[5] Alkin B, Fürst A, Schmid S, Gruber L, Holzleitner M and Brandstetter J 2024 Universal physics transformers: a framework for efficiently scaling neural operators *Advances in Neural Information Processing Systems* vol 37 pp 25152–94
[6] Lin Z *et al* 2023 Evolutionary-scale prediction of atomic-level protein structure with a language model *Science* **379** 1123–30
[7] Jumper J *et al* 2021 Highly accurate protein structure prediction with alphafold *Nature* **596** 583–9
[8] Huang K, Fu T, Gao W, Zhao Y, Roohani Y, Leskovec J, Coley C W, Xiao C, Sun J and Zitnik M 2022 Artificial intelligence foundation for therapeutic science *Nat. Chem. Biol.* **18** 1033–6
[9] Davies A *et al* 2021 Advancing mathematics by guiding human intuition with AI *Nature* **600** 70–74
[10] Zhu Q *et al* 2023 Automated synthesis of oxygen-producing catalysts from Martian meteorites by a robotic AI chemist *Nat. Synth.* **3** 1–10
[11] Degrave J *et al* 2022 Magnetic control of tokamak plasmas through deep reinforcement learning *Nature* **602** 414–9

[12] Szymanski N J *et al* 2023 An autonomous laboratory for the accelerated synthesis of novel materials *Nature* **624** 86–91
[13] Merchant A, Batzner S, Schoenholz S S, Aykol M, Cheon G and Cubuk E D 2023 Scaling deep learning for materials discovery *Nature* **624** 1–6
[14] Baum Z J, Yu X, Ayala P Y, Zhao Y, Watkins S P and Zhou Q 2021 Artificial intelligence in chemistry: current trends and future directions *J. Chem. Inf. Model.* **61** 3197–212
[15] Boiko D A, MacKnight R, Kline B and Gomes G 2023 Autonomous chemical research with large language models *Nature* **624** 570–8
[16] Bran A M, Cox S, White A D and Schwaller P 2023 Chemcrow: augmenting large-language models with chemistry tools (arXiv:2304.05376)
[17] Urbina F, Lentzos F, Invernizzi C and Ekins S 2022 Dual use of artificial-intelligence-powered drug discovery *Nat. Mach. Intell.* **4** 189–91
[18] Quach K 2022 Friendly AI chatbots will be designing bioweapons for criminals 'within years' (available at: www.theregister.com/2023/07/28/ai_senate_bioweapon/)
[19] Almeida A F, Moreira R and Rodrigues T 2019 Synthetic organic chemistry driven by artificial intelligence *Nat. Rev. Chem.* **3** 589–604
[20] Chen S and Jung Y 2021 Deep retrosynthetic reaction prediction using local reactivity and global attention *JACS Au* **1** 1612–20
[21] Pirie J 1958 The manufacture of hydrocyanic acid by the Andrussow process *Platinum Met. Rev.* **2** 7–11
[22] Yang L, Liu Y-T, Park Y, Park S-W and Chang S 2019 Ni-mediated generation of 'CN' unit from formamide and its catalysis in the cyanation reactions *ACS Catal.* **9** 3360–5
[23] Guo Z, Yi Y, Wang L, Yan J and Guo H 2018 Pt/TS-1 catalyst promoted C–N coupling reaction in CH_4–NH_3 plasma for HCN synthesis at low temperature *ACS Catal.* **8** 10219–24
[24] Yi Y *et al* 2021 Plasma-catalytic ammonia reforming of methane over Cu-based catalysts for the production of HCN and H_2 at reduced temperature *ACS Catal.* **11** 1765–73
[25] PubChem Classification Browser U.S. National Library of Medicine (available at: https://pubchem.ncbi.nlm.nih.gov/classification/#hid = 83)
[26] Ghasemi F, Mehridehnavi A, Pérez-Garrido A and Pérez-Sánchez H 2018 Neural network and deep-learning algorithms used in qsar studies: merits and drawbacks *Drug Discovery Today* **23** 1784–90
[27] Kumar V *et al* 2022 A decade of machine learning-based predictive models for human pharmacokinetics: advances and challenges *Drug Discovery Today* **27** 529–37
[28] Achiam J *et al* 2023 GPT-4 technical report (arXiv:2303.08774)
[29] Touvron H *et al* 2023 Llama 2: open foundation and fine-tuned chat models (arXiv:2307.09288)
[30] Bai Y *et al* 2022 Training a helpful and harmless assistant with reinforcement learning from human feedback (arXiv:2204.05862)
[31] Shen Y, Song K, Tan X, Li D, Lu W and Zhuang Y 2023 Hugginggpt: solving AI tasks with chatgpt and its friends in hugging face *Advances in Neural Information Processing Systems* vol 36 pp 38154–80
[32] Significant Gravitas: AutoGPT (available at: https://github.com/Significant-Gravitas/AutoGPT)
[33] Lu P, Peng B, Cheng H, Galley M, Chang K W, Wu Y N, Zhu S-C, Gao J 2023 Chameleon: plug-and-play compositional reasoning with large language (arXiv:2304.09842)
[34] AI4Science M R,Quantum M A 2023 The impact of large language models on scientific discovery: a preliminary study using GPT-4 (arXiv:2311.07361)
[35] He J *et al* 2024 SciMT-benchmark *GitHub SciMT/SciMT-benchmark* (available at: https://github.com/SciMT/SciMT-benchmark)
[36] Bhardwaj R and Poria S 2023 Red-teaming large language models using chain of utterances for safety-alignment (arXiv:2308.09662)
[37] Askell A *et al* 2021 A general language assistant as a laboratory for alignment (arXiv:2112.00861)
[38] Shen X, Chen Z, Backes M, Shen Y and Zhang Y 2024 Do anything now: characterizing and evaluating in-the-wild jailbreak prompts on large language models *Proc. 2024 on ACM SIGSAC Conf. on Computer and Communications Security* pp 1671–85
[39] Zheng L *et al* 2023 Judging LLM-as-a-judge with MT-bench and Chatbot Arena (arXiv:2306.05685)
[40] Schreck J S, Coley C W and Bishop K J 2019 Learning retrosynthetic planning through simulated experience *ACS Cent. Sci.* **5** 970–81
[41] Wan Y, Hsieh C-Y, Liao B and Zhang S 2022 Retroformer: pushing the limits of end-to-end retrosynthesis transformer *Int. Conf. on Machine Learning* pp 22475–90
[42] Xiong G *et al* 2021 ADMETlab 2.0: an integrated online platform for accurate and comprehensive predictions of ADMET properties *Nucl. Acids Res.* **49** 5–14
[43] Heid E, Greenman K P, Chung Y, Li S-C, Graff D E, Vermeire F H, Wu H, Green W H and McGill C J 2023 Chemprop: a machine learning package for chemical property prediction *J. Chem. Inf. Model.* **64** 9–17
[44] Sandbrink J B 2023 Artificial intelligence and biological misuse: differentiating risks of language models and biological design tools (arXiv:2306.13952)
[45] Acion L, Rajngewerc M, Randall G and Etcheverry L 2023 Generative AI poses ethical challenges for open science *Nat. Human Behav.* **7** 1–2
[46] Gopal A, Helm-Burger N, Justen L, Soice E H, Tzeng T, Jeyapragasan G,Grimm S, Mueller B and Esvelt K M 2023 Will releasing the weights of large language models grant widespread access to pandemic (arXiv:2310.18233)
[47] OpenAI: introducing superalignment 2023 (available at: https://openai.com/blog/introducing-superalignment)
[48] Anthropic: frontier threats red teaming for AI safety 2023 (available at: www.anthropic.com/index/frontier-threats-red-teaming-for-ai-safety)
[49] Gao K, Wang Y, Guan H, Wang Z, Pei Q, Hopcroft J E, He K and Wu L 2024 Tokenizing 3D molecule structure with quantized spherical coordinates (arXiv:2412.01564)
[50] Karpas E *et al* 2022 Mrkl systems: a modular, neuro-symbolic architecture that combines large language models, external knowledge sources and discrete reasoning (arXiv:2205.00445)
[51] Yao S, Zhao J, Yu D, Du N, Shafran I, Narasimhan K and Cao Y 2022 React: synergizing reasoning and acting in language models (arXiv:2210.03629)
[52] Wei J *et al* 2022 Chain-of-thought prompting elicits reasoning in large language models *Advances in Neural Information Processing Systems* vol 35 pp 24824–37
[53] Xie Y, Yi J, Shao J, Curl J, Lyu L, Chen Q, Xie X and Wu F 2023 Defending ChatGPT against jailbreak attack via self-reminders *Nat. Mach. Intell.* **5** 1486–96
[54] Singhal K *et al* 2023 Large language models encode clinical knowledge *Nature* **620** 172–80
[55] Sharma B, Chenthamarakshan V, Dhurandhar A, Pereira S, Hendler J A, Dordick J S and Das P 2023 Accurate clinical toxicity prediction using multi-task deep neural nets and contrastive molecular explanations *Sci. Rep.* **13** 4908
[56] Moor M *et al* 2023 Foundation models for generalist medical artificial intelligence *Nature* **616** 259–65

[57] U.S. Food and Drug Administration 2023 Artificial intelligence and machine learning (AI/ML) for Drug Development (available at: www.fda.gov/science-research/science-and-research-special-topics/artificial-intelligence-and-machine-learning-aiml-drug-development)

[58] Biden-Harris Administration 2023 FACT SHEET: Biden-Harris administration announces new actions to promote responsible AI innovation that protects Americans' rights and safety (available at: https://bidenwhitehouse.archives.gov/briefing-room/statements-releases/)

[59] Supermarket AI meal planner app suggests recipe that would create chlorine gas (available at: www.theguardian.com/world/2023/aug/10/pak-n-save-savey-meal-bot-ai-app-malfunction-recipes)

[60] Lowe D M 2012 Extraction of chemical structures and reactions from the literature *PhD Thesis* University of Cambridge

[61] Torkzadehmahani R, Nasirigerdeh R, Blumenthal D B, Kacprowski T, List M, Matschinske J, Spaeth J, Wenke N K and Baumbach J 2022 Privacy-preserving artificial intelligence techniques in biomedicine *Methods Inf. Med.* **61** 12–27

[62] Cirillo D *et al* 2020 Sex and gender differences and biases in artificial intelligence for biomedicine and healthcare *npj Dig. Med.* **3** 81

[63] Zirpoli C T 2023 Generative artificial intelligence and copyright law

[64] Lakomy M 2023 Artificial intelligence as a terrorism enabler? Understanding the potential impact of chatbots and image generators on online terrorist activities *Stud. Conflict Terrorism* **46** 1–21

[65] Huang L *et al* 2023 A survey on hallucination in large language models: principles, taxonomy, challenges and open questions (arXiv:2311.05232)

[66] Xue L and Bajorath J 2000 Molecular descriptors in chemoinformatics, computational combinatorial chemistry and virtual screening *Comb. Chem. High Throughput Screening* **3** 363–72

[67] Rogers D and Hahn M 2010 Extended-connectivity fingerprints *J. Chem. Inf. Model.* **50** 742–54

[68] Carhart R E, Smith D H and Venkataraghavan R 1985 Atom pairs as molecular features in structure-activity studies: definition and applications *J. Chem. Inf. Comput. Sci.* **25** 64–73

[69] Johnson M A, Maggiora G M 1990 Concepts and applications of molecular similarity

[70] Chemical weapons convention, annex on chemicals, B. schedules of chemicals 2023 (available at: www.opcw.org)

[71] PAN international list of highly hazardous pesticides 2023 (available at: https://pan-international.org/wp-content/uploads/PANHHPList.pdf)

[72] Schwaller P, Laino T, Gaudin T, Bolgar P, Hunter C A, Bekas C and Lee A A 2019 Molecular transformer: a model for uncertainty-calibrated chemical reaction prediction *ACS Cent. Sci.* **5** 1572–83

[73] Robey A, Wong E, Hassani H and Pappas G J 2023 SmoothLLM: defending large language models against jailbreaking attacks (arXiv:2310.03684)

[74] Hendrycks D, Burns C, Basart S, Zou A, Mazeika M, Song D and Steinhardt J 2020 Measuring massive multitask language understanding (arXiv:2009.03300)

[75] Li N *et al* 2024 The WMDP benchmark: measuring and reducing malicious use with unlearning (arXiv:2403.03218)